Jordan Thompson

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WORK EXPERIENCE

VRNDA Labs **June 2026 - Present**
Junior Mechatronics Engineer *Richmond, BC*

- Designed a minmum viable prototype for a modular syringe pump; from requirements to a CAD design with a bill of materials in 2 weeks.
- Engineered fluid delivery systems utilizing NEMA 17 stepper motors, CL42T-V41 drivers, and 3D printed parts designed in Fusion360; achieving 0.1–1200 mL/hr flow rates via microstepping at 51,200 steps/rev.

Kardium Inc **May 2025 – Aug 2025**
Biomedical Test Engineer Co-op *Burnaby, BC*

- Developed TOC and FT-IR test protocols to assess cleaning and sterilization of catheter-based ablation devices.
- Ensured ISO 10993 compliance and conducted non-parametric statistical analysis in Minitab, generating data to support FDA regulatory submissions.

Bioaugmentative Interfaces Lab **Nov 2023 - April 2025**
Mechantronics Co-op Student *Vancouver, BC*

- Developed a BLE profile on an nRF52840 chip using Zephyr RTOS to remotely control actuators and stream data from temperature and strain sensors.
- Designed an enlarged proof of concept PCB utilizing a ublox BMD340 to validate electrical performance for migration to a minutrized PCB for implantion into rodents.
- Built a C++/Arduino-based phantom bladder rig with a 3D-printed stand and flow meter to validate flow rates and actuator force, subsequently supporting in-vivo animal trials.

Undergraduate Research Assistant *Vancouver, BC*

- Optimized a shape-memory alloy actuator's duty cycle, achieving 78% liquid voiding efficiency while restricting temperature increases to <2°C.
- Designed and iterated upon modular 3D-printed molds in OnShape to streamline actuator insulation, minimize air bubbles, and improve manufacturing repeatability.

SKILLS

Programming Languages: C/C++, MATLAB, Python
Microcontrollers and Communication Protocols: Arduino, ESP32, Nordic nrf52840, BLE, Zephyr RTOS
Electronics and PCB: Altium, Eagle, Oscilloscope, Digital Multimeter, Soldering
CAD and Rapid Prototyping: SolidWorks, Onshape, Fusion360, 3D Printing (FDM & MSLA), Laser Cutting

PROJECTS

Dr. Wobbles, Open Source Pediatric Pain Relief Device - Website **2025 - 2026**

- Designed a screwless, modular 3D-printed housing in OnShape, enabling intuitive non-technical user assembly in under 13 minutes for an open-source medical device that provides pain relief through vibration and cooling.
- Engineered a motor-driving PCB that delivers stable varying vibrational frequency modes (68–172 Hz) with a brushed DC motor, using noise filtering techniques to prevent MCU brownouts (lowpass filter + flyback diode).

Spatio-Temporal Electromyography Data Analysis Algorithm **2025 - 2026**

- Built an automated TKEO-based detection algorithm to successfully isolate transient signal features in low signal-to-noise environments.
- Processed 64-channel continuous electromyography sensor data using an 8th-order Butterworth and 60 Hz notch filters to eliminate baseline and powerline noise.
- Validated spatial displacement predictions against 3D-scanned physical coordinates, significantly reducing euclidean distance errors across dynamic tests.

EDUCATION

University of British Columbia **Graduated April 2026**
BASc, *Biomedical Engineering specialized in Mechatronics*: 86% Overall Average, 92% 300-400 Course Level Average